Data Analytics 2025-2026

IT ACADEMY

Healthcare Data Analysis: Patient, Treatment, and Cost Insights

Exploratory Data Analysis and Prescription Text Processing

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Agenda

- Project Motivation & Goals
- Datasets Overview
- Methodology
  - Exploratory Data Analysis (EDA)
  - Unstructured Text Processing
- Key Findings
- Conclusions
- Future Work

Project Motivation

- Healthcare systems generate large amounts of patient and treatment data
- Understanding patterns can help:
  - Improve resource allocation
  - Identify cost drivers
  - Analyze treatment effectiveness

WHY THIS PROJECT?

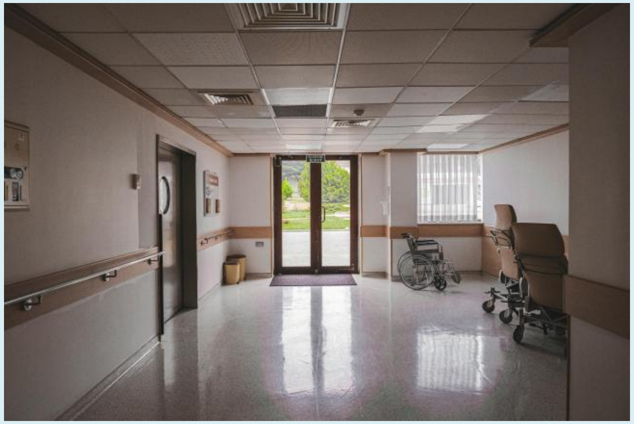
- **1st Objective: Dataset for analysis**

mockaroo

https://www.kaggle.com/

Project Goal

- Extract meaningful insights about COSTS from healthcare data
- Identify cost drivers



Datasets Overview

https://www.kaggle.com/

HealthcareDataset1 (Patients info)

- PatientID,
- PatientName
- Age
- Gender
- Bloodtype
- Diagnosis
- Treatment
- AdmissionDate
- DischargeDate
- TotalBill ($)
- Full Prescription Details (unstructured)

CLINICAL INFORMATION

1000 rows, 11 columns

HealthcareDataset2 (Hospital info)

- PatientID
- Hospital
- DoctorName
- RoomNumber
- DailyCost (S)
- TreatmentType
- RecoveryRating

OPERATIONAL INFORMATION

1000 rows, 7 columns

Methodology

Exploratory Data Analysis

Data Quality Check, Cleaning & Data Transformation

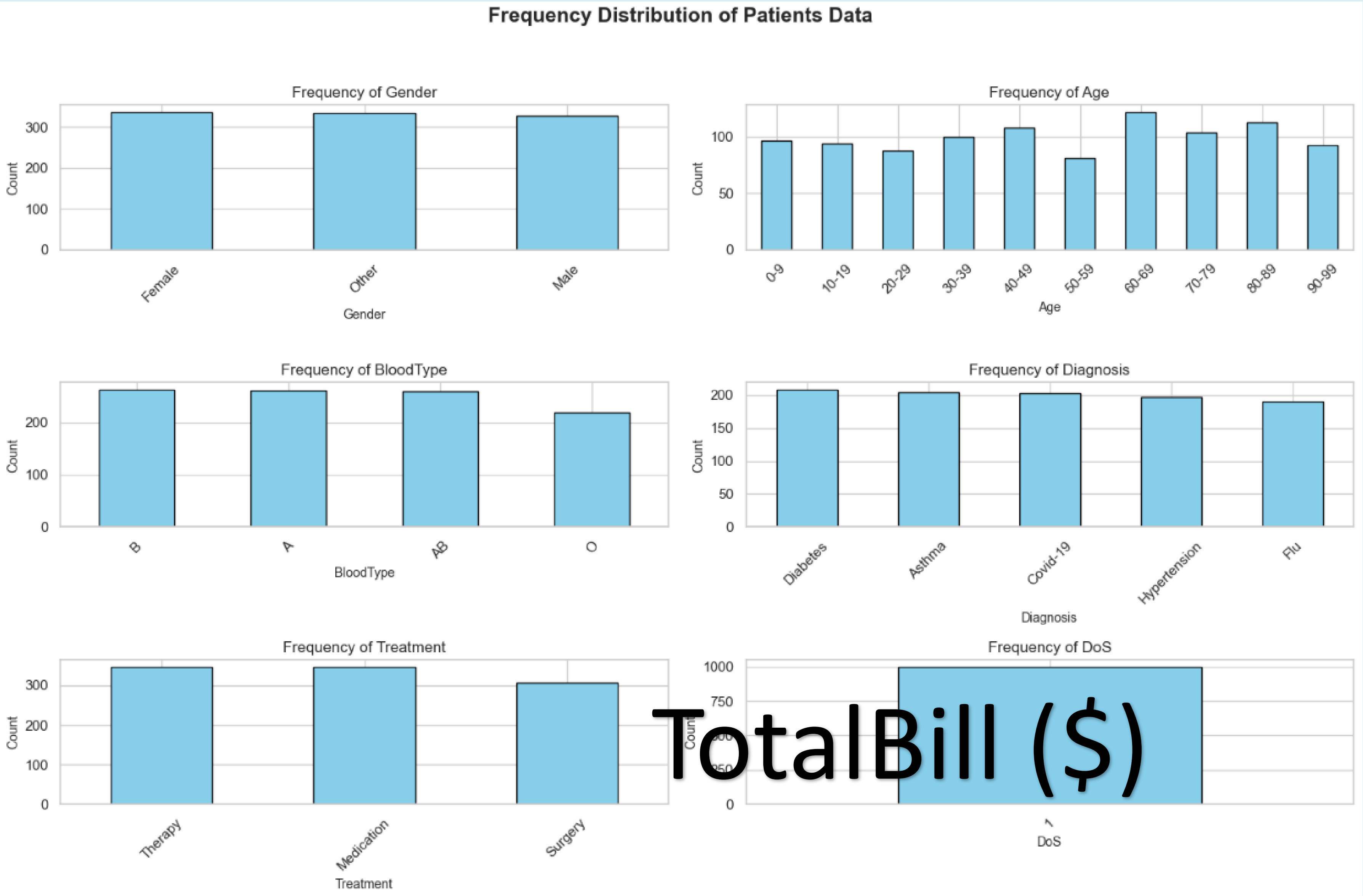
- Date consistencies
- Identify Nulls
  - Patients (6% TotalBill)
  - Doctors (7,8% RecoveryRate)
- Identify Duplicates
- Identify Outliers
- Extract information from the "Full Prescription Details" (patients)

- DoS (Days of Stay)
- Temporary Analysis
  - AdmissionDate
    - AD_Year
    - AD_Quarter
    - AD_Month
    - AD_Day
  - DischargeDate
    - DD_Year
    - DD_Quarter
    - DD_Month
    - DD_Day

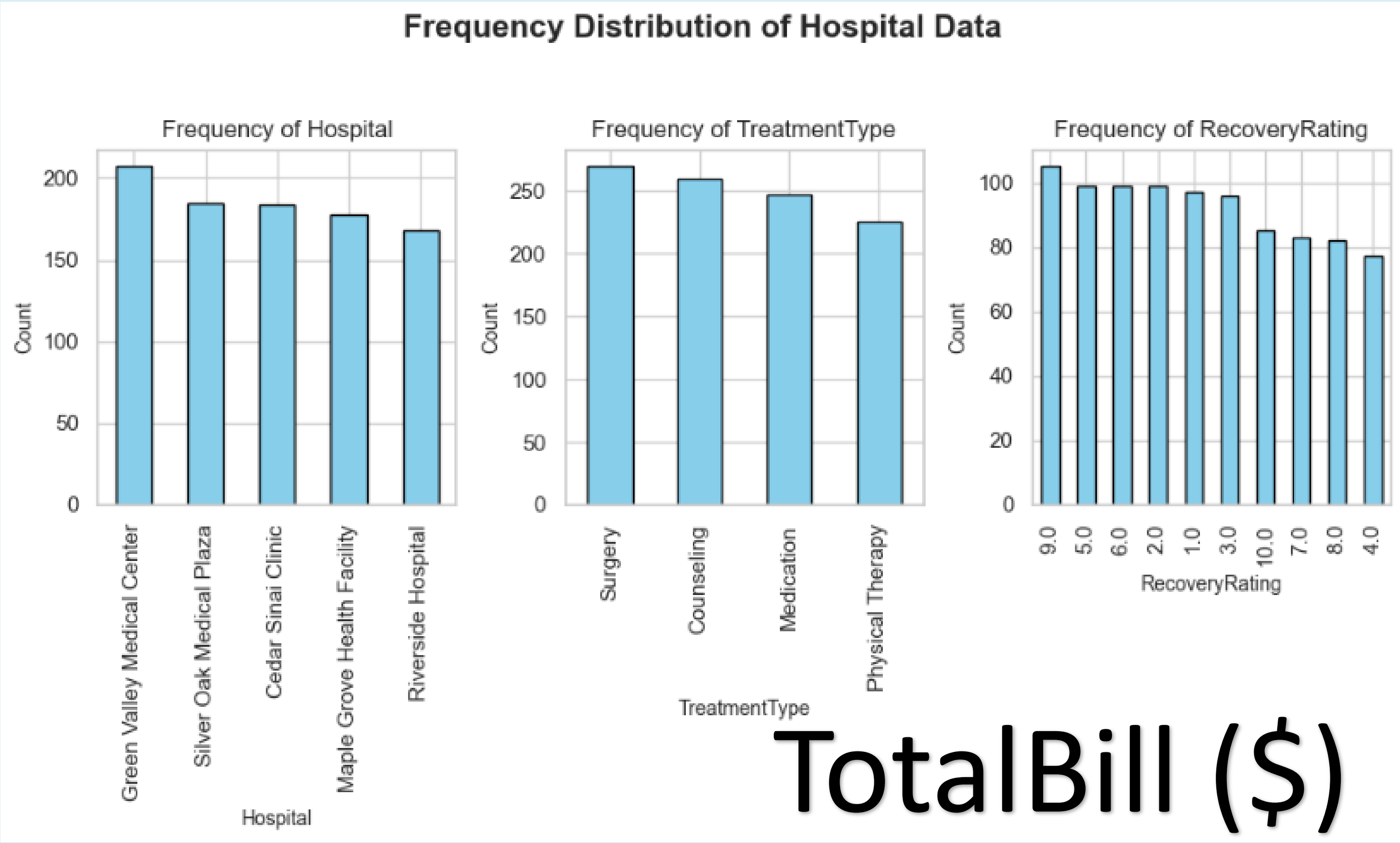
- Combined dataset (PatientID):

df = clinical information +hospital operational data (# 1000 rows, 32 variables)

Patients Breakdown



Hospital Breakdown

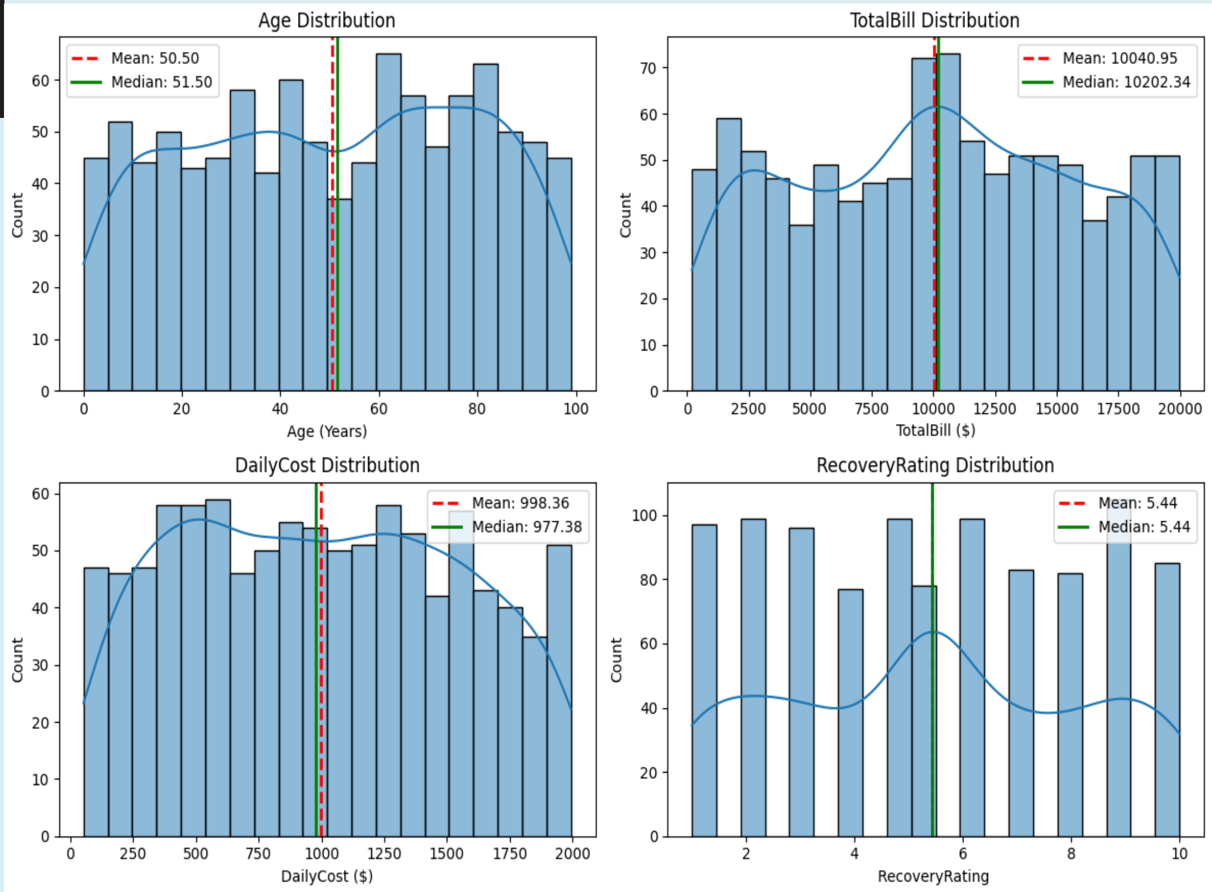


TotalBill ($)

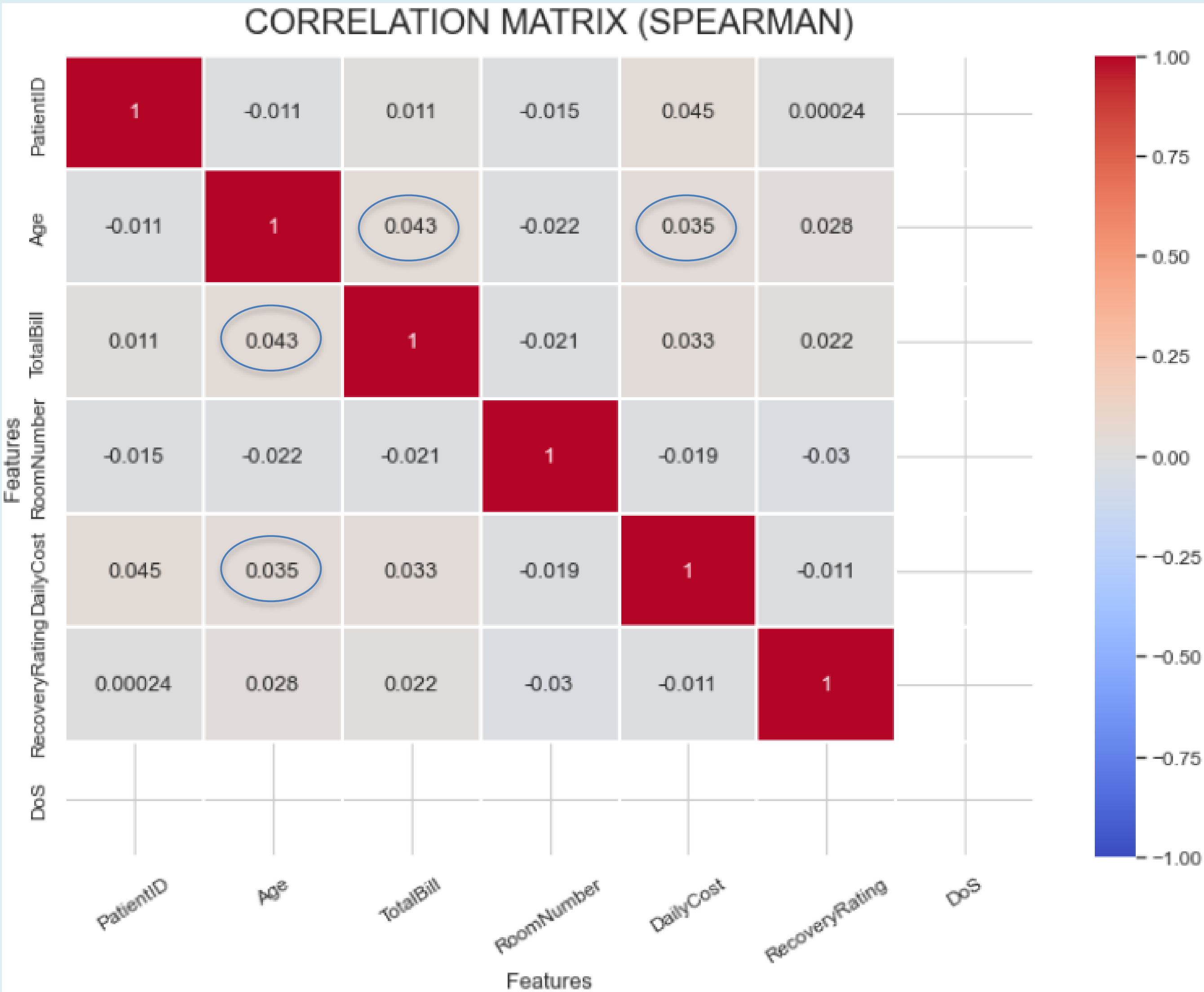
Descriptive Statistics & Variable Distribution

	Age	TotalBill	DailyCost	RecoveryRating	DoS
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.0
mean	50.500000	10040.947498	998.358349	5.436009	1.0
std	28.599859	5626.658910	545.178933	2.776767	0.0
min	0.000000	200.928022	52.023698	1.000000	1.0
25%	26.000000	5326.999625	527.573516	3.000000	1.0
50%	51.500000	10202.339277	977.376355	5.436009	1.0
75%	75.000000	14552.941322	1459.068550	8.000000	1.0
max	99.000000	19979.201527	1994.047594	10.000000	1.0

Normal Distributions????



Quantitative Variables Correlation



Statistical Test Results
Significant Differences and Associations

(T-test and Chi²)

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Division variable: Age
T-statistic: -2.280, p-value: 0.023
→ There is a statistically significant difference in 'TotalBill' between the two groups based on the median of the variable.

Division variable: DailyCost
T-statistic: -0.828, p-value: 0.408
→ No significant difference in 'TotalBill' between the groups based on the median of the variable.

Division variable: RecoveryRating
T-statistic: -0.147, p-value: 0.883
→ No significant difference in 'TotalBill' between the groups based on the median of the variable.
```

```
Variable: Gender
Chi2: 1915.7667, p-value: 0.4330
→ No significant relationship with TotalBill

Variable: BloodType
Chi2: 2860.0548, p-value: 0.4909
→ No significant relationship with TotalBill

Variable: Diagnosis
Chi2: 4000.0000, p-value: 0.0167
→ There is a significant relationship with TotalBill

Variable: Treatment
Chi2: 2000.0000, p-value: 0.0657
→ No significant relationship with TotalBill

Variable: Hospital
Chi2: 3527.6197, p-value: 0.4986
→ No significant relationship with TotalBill

Variable: DoctorName
Chi2: 350587.0304, p-value: 0.5552
→ No significant relationship with TotalBill

Variable: TreatmentType
Chi2: 2863.2094, p-value: 0.4743
→ No significant relationship with TotalBill
```

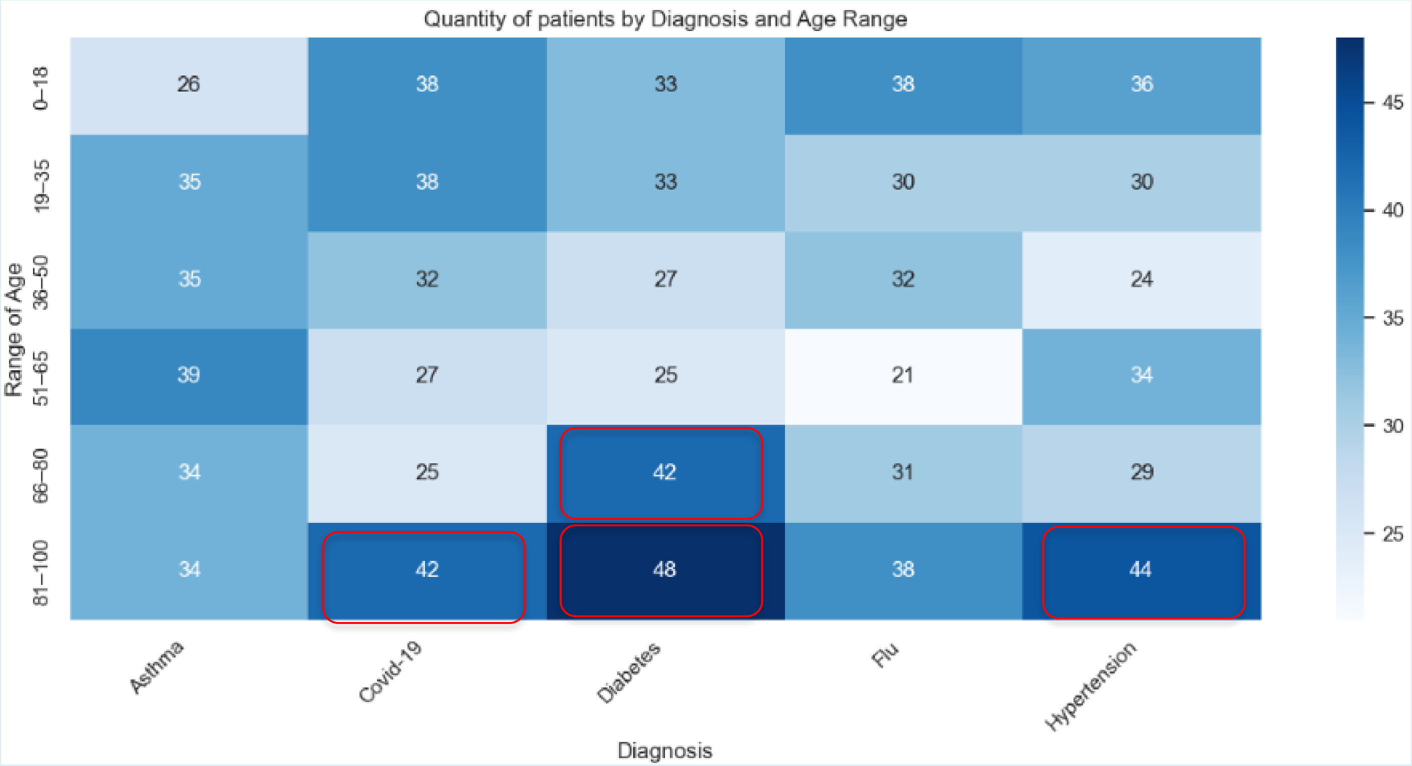
```
Variable: Medicine1
Chi2: 27663.8823, p-value: 0.4534
→ No significant relationship with TotalBill

Variable: Medicine2
Chi2: 27652.0842, p-value: 0.4733
→ No significant relationship with TotalBill

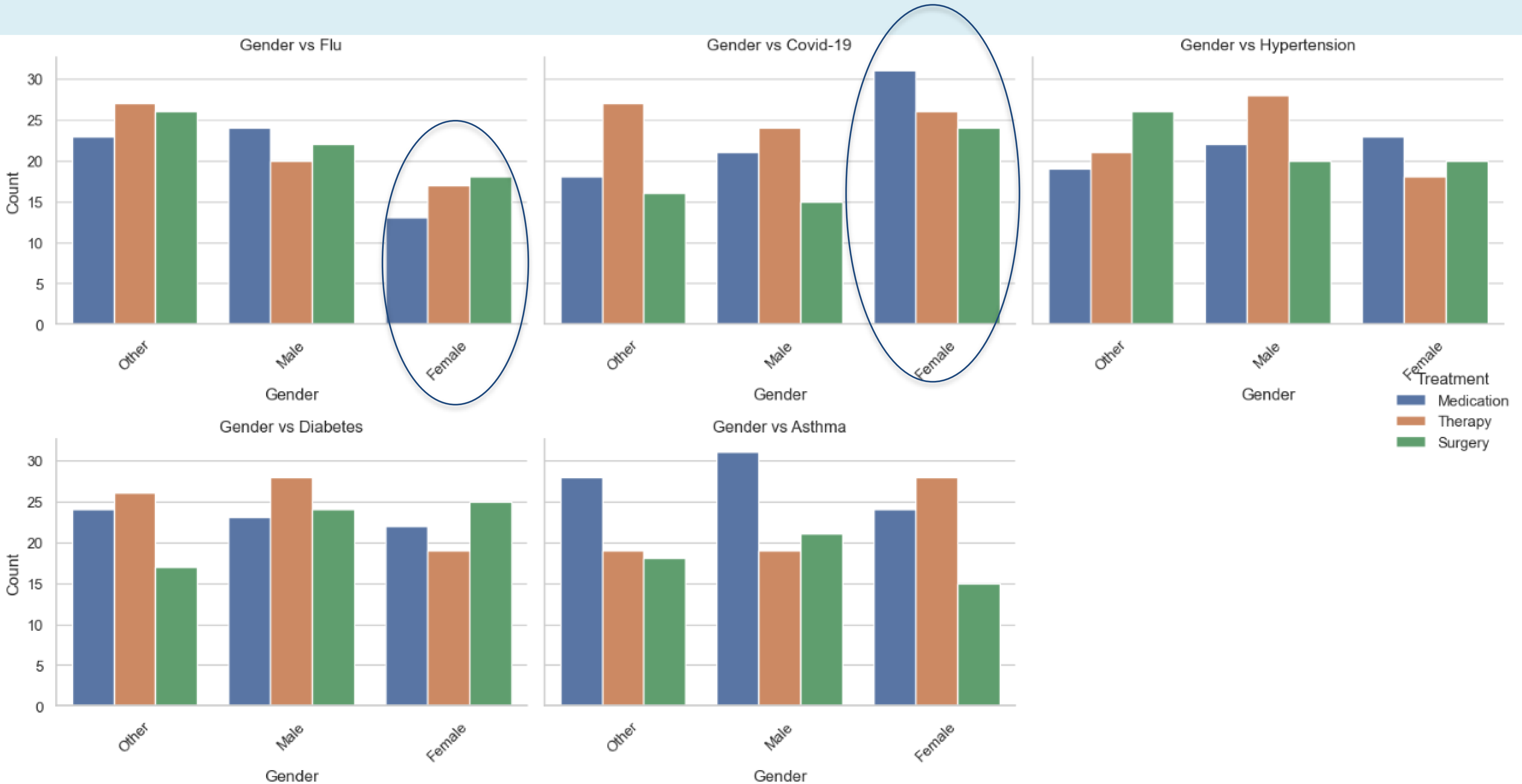
Variable: Medicine3
Chi2: 18929.1735, p-value: 0.4553
→ No significant relationship with TotalBill

Variable: Medicine4
Chi2: 10147.4854, p-value: 0.4244
→ No significant relationship with TotalBill
```

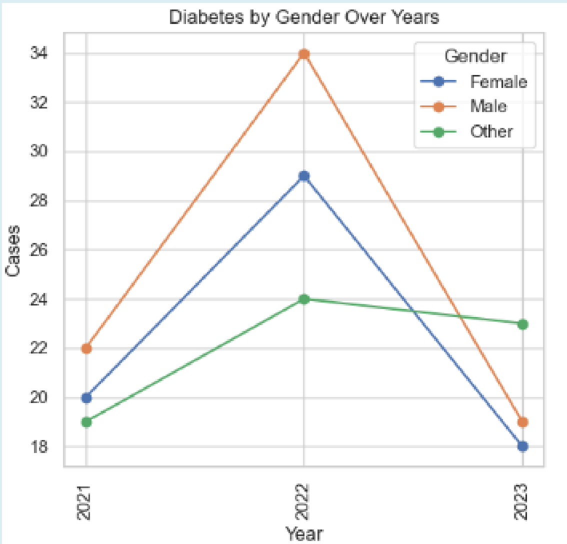
Key Findings: Diagnosis per Age

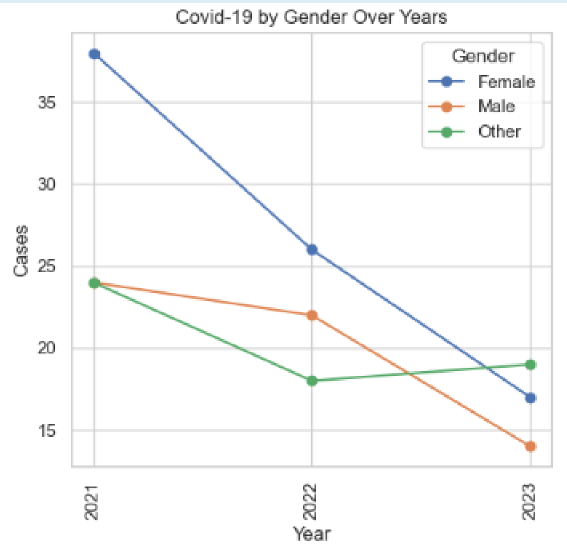


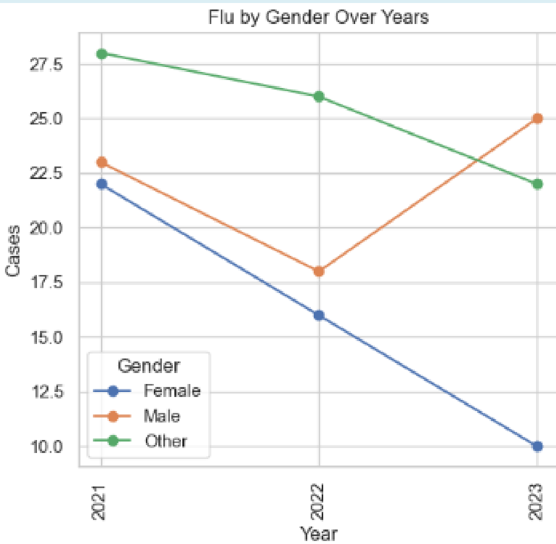
Key Findings: Diagnosis per Gender

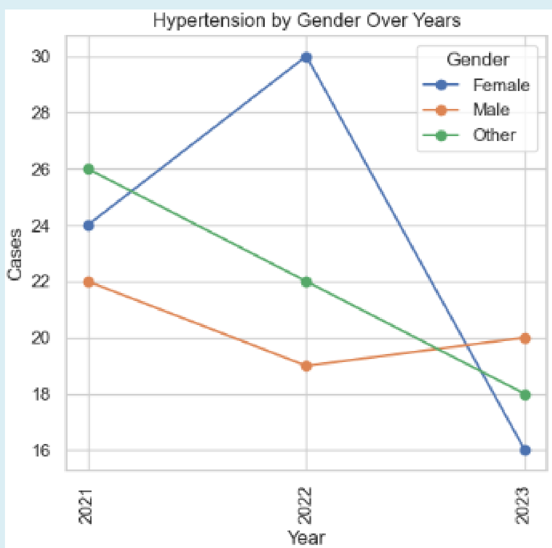


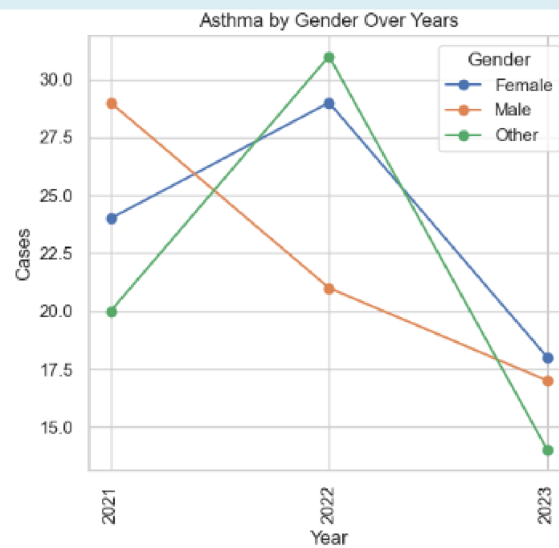
Key Findings: Diagnosis by gender over years



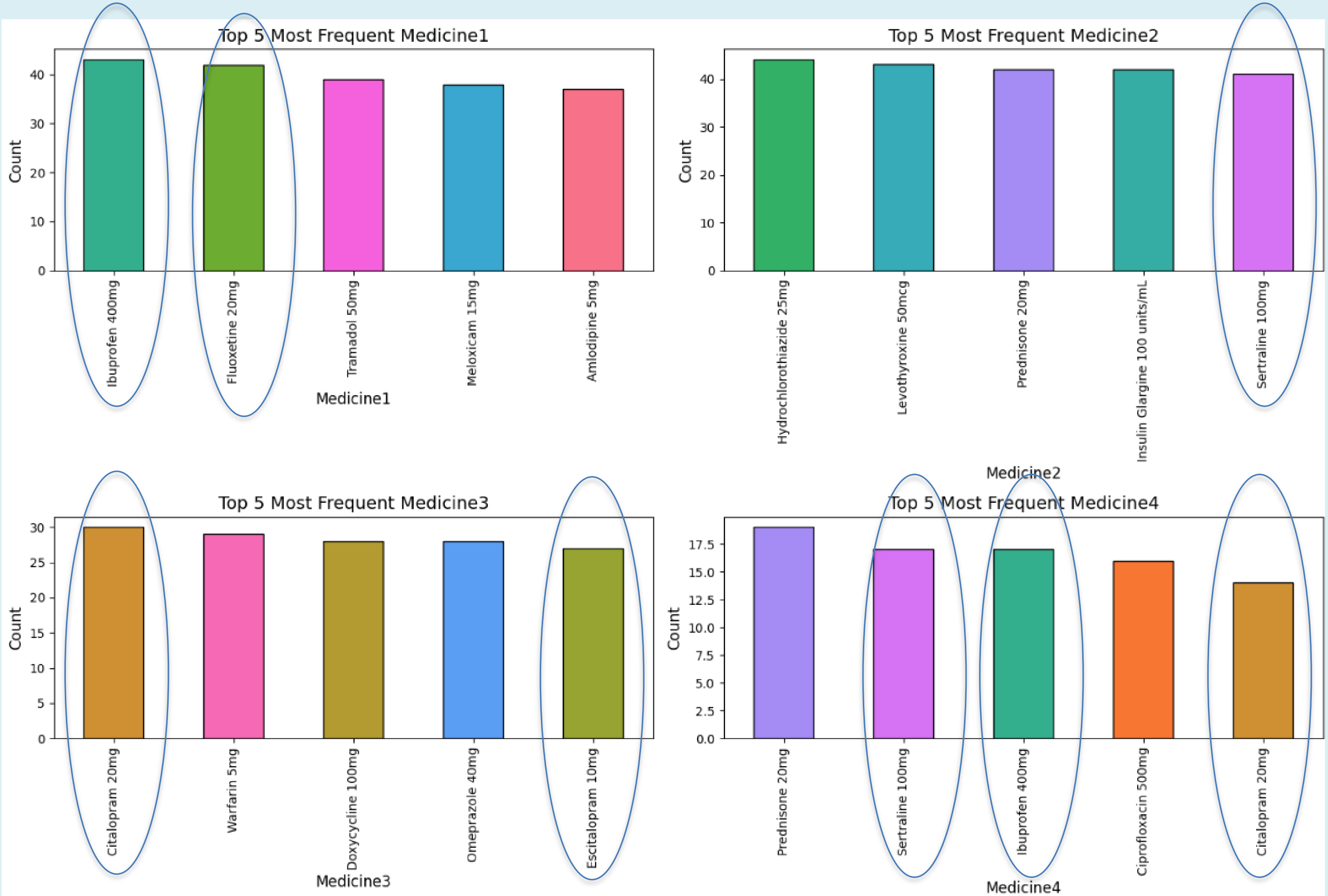




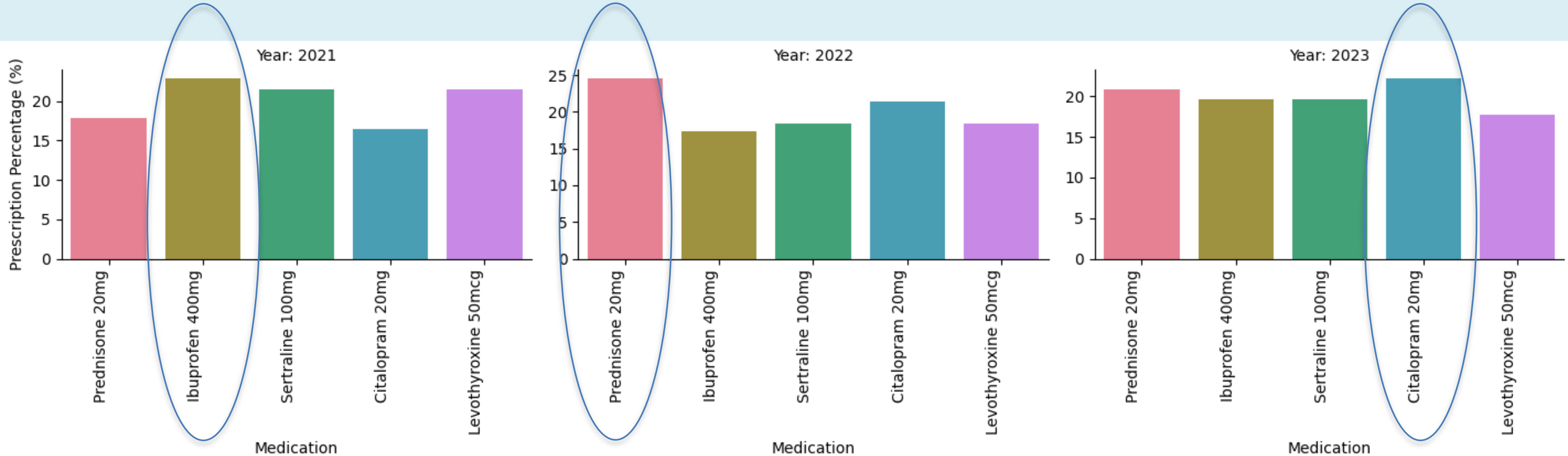




Key Findings: Top 5 Most Frequent Medicines

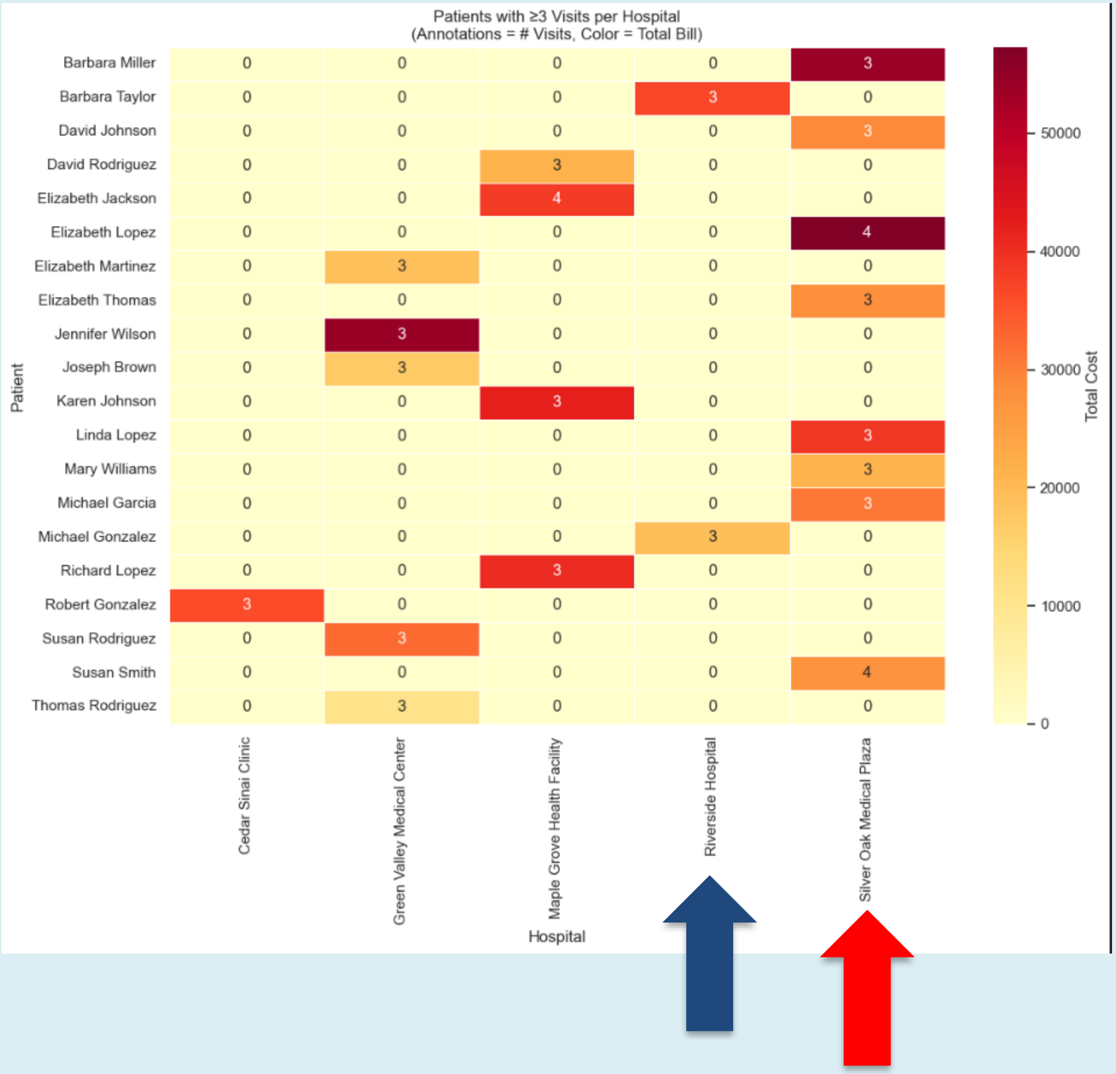


Key Findings: Top 5 Most Frequent Medecines (over years)



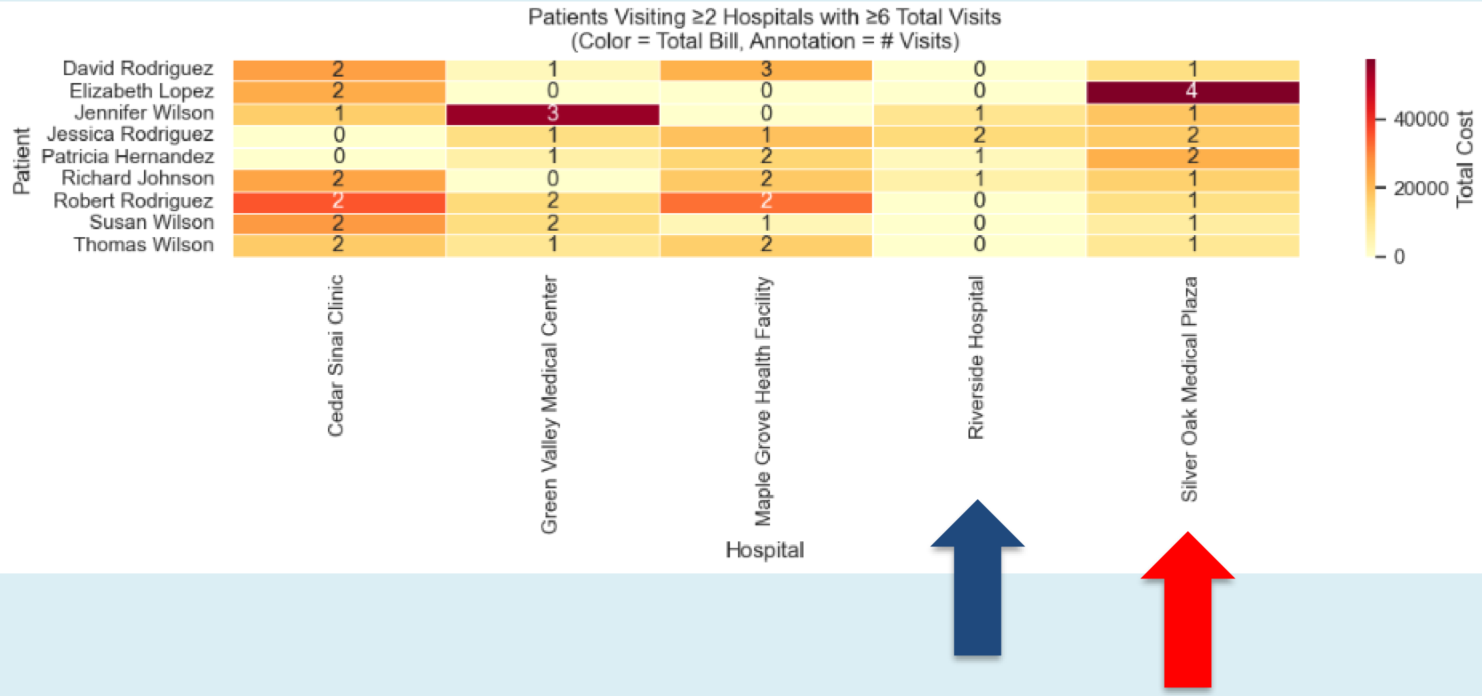
YEAR 2021 – Ibuprofen 400 mg
YEAR 2022 – Prednisone 20 mg
YEAR 2023 – Citalopram 20 mg

Key Findings: Patients with recurrence (I)



Patient	Cedar Sinai Clinic	Green Valley Medical Center	Maple Grove Health Facility	Riverside Hospital	Silver Oak Medical Plaza
Barbara Miller	0	0	0	0	3
Barbara Taylor	0	0	0	3	0
David Johnson	0	0	0	0	3
David Rodriguez	0	0	3	0	0
Elizabeth Jackson	0	0	4	0	0
Elizabeth Lopez	0	0	0	0	4
Elizabeth Martinez	0	3	0	0	0
Elizabeth Thomas	0	0	0	0	3
Jennifer Wilson	0	3	0	0	0
Joseph Brown	0	3	0	0	0
Karen Johnson	0	0	3	0	0
Linda Lopez	0	0	0	0	3
Mary Williams	0	0	0	0	3
Michael Garcia	0	0	0	0	3
Michael Gonzalez	0	0	0	3	0
Richard Lopez	0	0	3	0	0
Robert Gonzalez	3	0	0	0	0
Susan Rodriguez	0	3	0	0	0
Susan Smith	0	0	0	0	4
Thomas Rodriguez	0	3	0	0	0

Key Findings: Patients with recurrence (II)



Patient	Cedar Sinai Clinic	Green Valley Medical Center	Maple Grove Health Facility	Riverside Hospital	Silver Oak Medical Plaza
David Rodriguez	2	1	3	0	1
Elizabeth Lopez	2	0	0	0	4
Jennifer Wilson	1	3	0	1	1
Jessica Rodriguez	0	1	1	2	2
Patricia Hernandez	0	1	2	1	2
Richard Johnson	2	0	2	1	1
Robert Rodriguez	2	2	2	0	1
Susan Wilson	2	2	1	0	1
Thomas Wilson	2	1	2	0	1

David Rodriguez – 7 visits in 4 hospitals
Robert Rodriguez – 7 visits in 4 hospitals

Conclusions

- Successful Data Cleaning & Transformation
- Basic Text Information Extraction (“Full prescription Details”)
- Successful Exploratory Data Analysis
- Poor Identification of Clinical vs Economic Insights
  - Most variables do not show significant relationship with cost
  - Only Age and Diagnosis exhibit statistically relevant differences
  - Elderly patients with chronic diseases account for the majority of hospital visits
  - Preliminary analysis shows a decline in hospital visits in 2023
  - Steady growth observed in mental health medication prescriptions
  - Patient repeat visits vary by hospital
- Main limitation is the synthetic nature of dataset

Future Work

- Apply Machine Learning Models to:
  - Predict TotalBill
  - Predict RecoveryRating
- Use more advanced NLP techniques for
  - Prescription analysis
- Include additional variables:
  - Comorbidities (ie. Diabetes, Hypertension)
  - Laboratory results (i.e glucose, creatinine…)
- Validate findings on real-world healthcare datasets

Thank you

"You lose your curiosity when you stop learning"

Katherine Johnson- Physicist, Space Scientist and Mathematician